

Framing Fairness: News Narratives and the Politics of Welfare

Katherine Ren
Adviser: Mihir Kshirsagar

Abstract

In this paper, I use a computational approach to analyze the differences in framing and sentiment that Fox News uses when talking about federal welfare programs such as TANF, Social Security, and food stamps versus the framing and sentiment used when talking about COVID stimulus checks. My aim is to investigate these differences and make recommendations to policymakers in order to shift conservative framings of federal welfare programs to support a more accessible, direct-payment welfare program (similar in function to COVID stimulus checks). In order to get a nuanced view of the framing and sentiment differences, I analyzed selected articles from Fox News using sentiment analysis, frequency analysis, and framing analyses. I find that COVID stimulus checks are overwhelmingly spoken of using an economic centered frame, while federal welfare programs tend to favor narratives that center the immorality of welfare recipients, pushing forward the ideas that tax dollars should not be invested into welfare programs and that welfare should be less accessible in order to weed out undeserving populations. In order to reconcile these differences for my primary political aim of welfare reforms that make financial help more accessible, I tentatively recommend that politicians begin speaking of welfare programs, not as a moral issue, but as an economic issue – that poverty hurts everyone, not just the poor, and in turn, eliminating poverty helps everyone and stimulates the economy.

1. Introduction

The US Government's treatment of poor Americans is uniquely bad on a global scale, especially among other developed countries.^[21] There exists a distinct American stigma against receiving government assistance through welfare programs, which manifests as a general disapproval of such

programs. This then leads to welfare reform, usually in the form of cuts and extra restrictions which only serve to harm the poor instead of helping them.[41]

The portrayal of poor Americans in media, specifically political news media, is a driving force for welfare reforms that further oppress our nation's most vulnerable. In their 2013 paper, "Framing the Poor: Media Coverage and US Poverty Policy, 1960-2008", Rose and Baumgartner find that the relative generosity of US government policy is highly related to framings of the poor in newspaper stories. For instance, more generous policies were spurred on and accompanied by framings of the poor as unlucky, but deserving of help, and harsher policies followed news coverage of the poor as lazy and undeserving. Rhetoric manifests in real-world policy changes. [41]

In this paper, I hope to investigate better ways of advocating for less restrictive, more generous welfare programs using COVID-19 stimulus checks as a case study. I take a computational approach to investigating this question, using a sentiment analysis model as well as frequency analyses and framing dictionaries in order to properly analyze a large corpus of data. After all, these stimulus checks were passed by a Republican president, and were generally approved of by Republicans, especially years after they were distributed.[34] These checks had no restrictions on what the money could be spent on and did not require privacy-violating and humiliating processes to obtain, unlike many of our largest welfare programs. Furthermore, they had an immensely successful effect on the struggling US economy and lowered poverty rates – the usual goals of welfare programs. [30] How do the ways we talk about welfare programs compare to the ways that we talked about COVID stimulus checks, and what framings that worked for COVID stimulus checks can we consider applying to welfare programs, especially as we've continued to observe their success?

2. Background

2.1. Welfare in the USA: A Brief History of Moral Framings of Help

Among the world's 26 most developed countries, the US has the highest poverty rate — about one in five children and one in ten adults live in poverty. Impoverished Americans are overexposed to violent crime and tend to have less access to high quality education, healthcare, and healthy food.

[21] Poverty is not just pitiful; it's fatal. Experts find that poverty kills as many people as dementia, diabetes and Alzheimer's. [40]

Politicians across the isle want to reduce poverty, yet no one wants to associate themselves with welfare programs like SSI (Supplemental Security Income), or TANF (Temporary Assistance for Needy Families). "Welfare," as Copeland writes, "is an orphan." [31] Hated by conservatives for being handouts for the undeserving, hated by liberals for being ineffective, welfare is a policy no one wants to claim, yet nobody dares to kill completely.

The two narratives diverge primarily in their moral framing of poverty as a state. Conservatives tend to see poverty as a personal failing, and thus perceive welfare programs meant to help the impoverished as giving unnecessary help to those who are simply receiving their comeuppance. Liberals tend to see poverty as an effect of a person's environment, not their character, and see government welfare programs as ineffective for treating the root causes of poverty. [23] The result

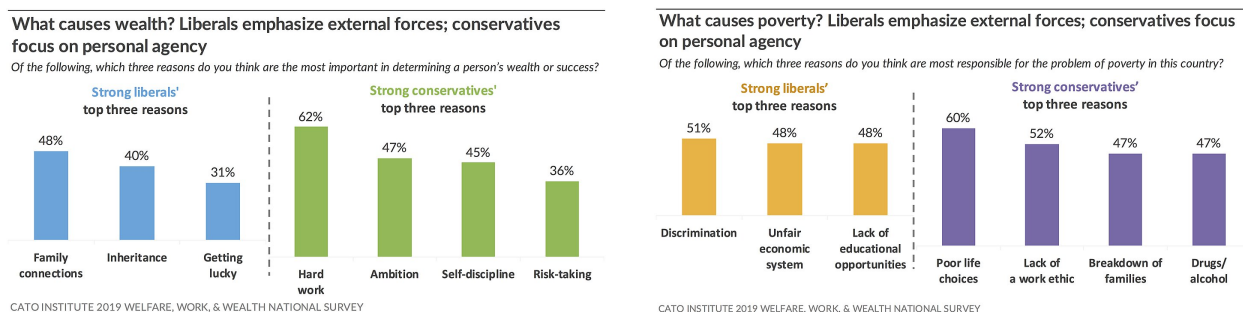


Figure 1: Liberal and Conservative Views of Causes of Wealth/Poverty, from the Cato 2019 Survey [23]

of these differences in narrative is an ideologically orphaned welfare state that exists but is neither generous nor socially transformative: the worst of both worlds. [31]

In order to properly explore and imagine a meaningfully different framing from the welfare state narratives of the past, I find it indispensable to study how and when these divergent views of poverty and welfare developed in the United States. What were the cultural zeitgeists that birthed and supported these narratives, and what legacy have they left behind in our modern framings of welfare?

2.1.1. President Johnson: Poverty as the “Enemy”

President Lyndon B. Johnson’s “War on Poverty” from 1964 to 1968 greatly expanded social safety nets for the poor, expanding access to federal welfare programs, education, and jobs. However, these expansions occurred while the United States was increasingly involved in the Vietnam War, which limited the economic resources available for reallocation. Thus, almost all of Johnson’s reforms were met with resistance, particularly from conservatives who thought that the limited amount of federal money available should not be used to help the poor. Still, Johnson’s rhetoric around poverty made it into written law and remained in the American political consciousness as a historic quote:

On similar occasions in the past we have often been called upon to wage war against foreign enemies which threatened our freedom. Today we are asked to declare war on a domestic enemy which threatens the strength of our nation and the welfare of our people. If we now move forward against this enemy—if we can bring to the challenges of peace the same determination and strength which has brought us victory in war—then this day and this Congress will have won a secure and honorable place in the history of the nation and the enduring gratitude of generations of Americans yet to come.

-President Lyndon B. Johnson

Johnson framed “poverty” as the common enemy, not “the poor.” In this, he recognized poverty as a social force that a community needed to quell by fighting alongside and supporting each other, not an innate consequence of individual actions. He equates the “strength of our nation” with the “welfare of our people,” framing welfare as something we need to protect and nurture in order to pursue “victory” against poverty[22].

2.1.2. Reagan and M&Ms (Murray and Mead): Welfare Programs as a Trap

Even before he came into office, Reagan was an outspoken critic of Johnson’s reforms — he argued that federally funded social programs attempted to control people using force and coercion [17]. When he eventually became president in 1981, his administration made a concerted effort to reduce federal welfare program spending[33].

One of the oft-cited inspirations of the Reagan administration's welfare cuts was *Losing Ground: American Social Policy, 1950-1980*, a book by Charles Murray that argued that welfare programs designed to reduce poverty actually increased it [16]. He claimed that the only corrective that would prevent and reduce poverty was the harsh reality of the free marketplace — if welfare programs did not exist, the fear of poverty would loom larger and prevent citizens from making decisions that will make them poor. That is, that the mere existence of welfare programs creates an implicit dependency on them [31], a sentiment that Reagan echoes in his State of the Union Message, where he called welfare programs a “spider’s web of dependency” [17]. This is a conservative and libertarian framing of welfare programs, focused on the self-governance of citizens, thereby attributing poverty entirely to the individual’s own choices and responsibilities.

As this framing became more popular, characterizations of welfare recipients emerged. For instance, the famous myth of the “welfare queen,” an image of a welfare recipient as a poor, unmarried Black mother who lived lavishly on welfare funds while committing crimes and doing drugs, appeared frequently in Reagan’s speeches in order to rally support for his federal welfare program cuts [17]. By mapping the character of welfare onto a Black woman’s body, he also successfully connected the act of receiving welfare to existing racist and sexist caricatures, further solidifying images of welfare recipients as greedy and lazy cheaters. Other Reaganisms for welfare recipients included “welfare chiselers” and “poverty pimps,” which further strengthened the perception of ties between welfare and crime [17].

As these characterizations of welfare recipients became popularized, a different but related conservative framing began to emerge. During Reagan’s second term, in 1986, Lawrence M. Mead wrote *Beyond Entitlement: The Social Obligations of Citizenship*, which would form a basis for a new era of welfare [31]. Mead argued that “most [poverty] is due simply to the fact that [the poor] are not employed,” and elaborated that “this failure to work cannot be explained by any barrier outside the poor themselves” [13] (This view of poverty is almost identical to JD Vance’s argument in *Hillbilly Elegy* — though Mead’s name may not be widely known, he has made an enduring impression on American perceptions of the poor [11].) He believed that a strict adherence

to conservative family values would eliminate poverty.

Thus, his ideal vision of welfare involved tying benefits closely to what he saw as fulfilling the five duties of the American Citizen:

1. Working if one is able
2. Supporting one's family as much as possible
3. Speaking and Writing English
4. Learning enough to get a job
5. Following the law

Mead's welfare system is highly conditional and authoritative, and the rhetoric he uses to push forward his argument focuses largely on social and moral integrity and the duties of an American citizen [31].

2.1.3. Clinton: "Ending Welfare as We Know It"

Much of Mead's vision for a conditional welfare system came into fruition in the passing of the Personal Responsibility and Work Opportunity Reconciliation Act of 1996 (PRWORA), which transformed the Aid to Families with Dependent Children (AFDC) program with the Temporary Assistance for Needy Families (TANF) Program. Though the authors of the rhetoric and arguments that pushed forward this reform (Reagan, Murray, Mead) were conservatives, the fatal blow to AFDC was dealt by Democratic president Bill Clinton, who signed PRWORA into law. Clinton ran on a platform that promised to "end welfare as we know it," appealing to both Republican and Democratic voters who had been convinced by Reagan-era narratives that highlighted the "culture of poverty" and cases of welfare fraud [2].

[PRWORA] gives us a chance we haven't had before to break the cycle of dependency that has existed for millions and millions of our fellow citizens, exiling them from the world of work. It gives structure, meaning and dignity to most of our lives.

- Clinton in 1996 [2]

Despite Republicans often attacking the Democratic Party as the welfare-state party that gives handouts to the undeserving poor, the Democratic Party's rhetoric and actions regarding welfare

have not meaningfully diverged from conservative rhetoric and actions since Johnson’s presidency. TANF made welfare significantly more conditional than AFDC, now including work-requirements and time-limits for welfare-recipients[27].

Feature	AFDC	TANF
Program Start	1935 (New Deal)	1996 (Welfare Reform Act)
Funding	Federal entitlement program	Federal block grant to states
Work Requirements	No federal work requirements	Requires recipients to work or participate in job preparation activities
Time Limits	No time limit on benefits	60-month lifetime limit (may vary by state)
State Flexibility	Uniform federal standards	States have significant discretion in design and implementation
Goal	Income support for low-income families with children	Promote work, reduce dependency, and support family formation

Table 1: Key Differences Between AFDC and TANF [27]

The bipartisan shift from AFDC to TANF, driven in large part by growing public disapproval of welfare recipients, demonstrates how rhetoric—particularly when it originates from authoritative sources such as the president—can shape public opinion and ultimately produce significant policy change.

Number in Poverty and Poverty Rate: 1959 to 2019

(Population as of March of the following year)

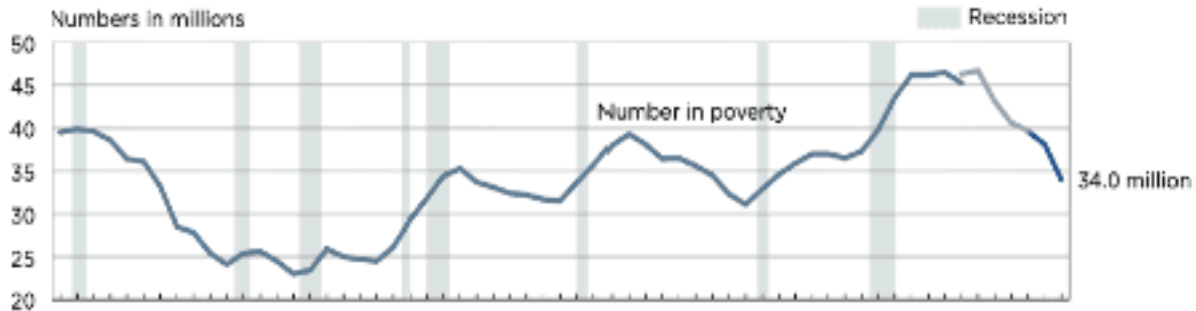
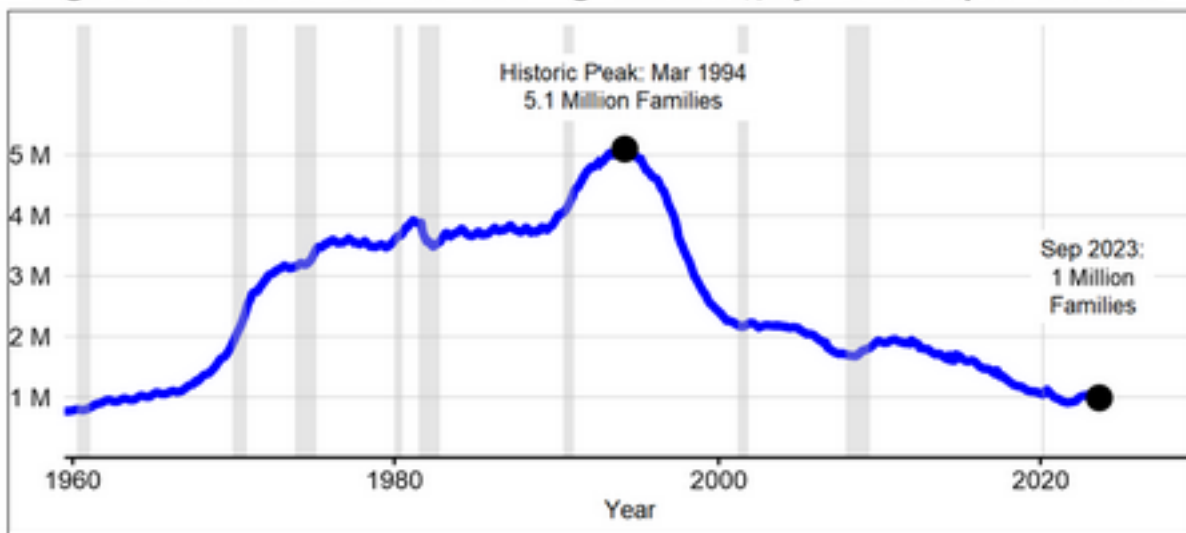


Figure 2. Number of Families Receiving Assistance, July 1959 to September 2023



Source: Congressional Research Service (CRS), with data from the U.S. Department of Health and Human Services (HHS).

Notes: Shaded areas indicate periods of economic recession. Since FY2000, data include families receiving assistance in separate state programs, funded with state dollars and accounted for separately but with spending counted toward the TANF maintenance of effort (MOE) requirement.

Figure 2: Number of Families Receiving Welfare vs Number in Poverty [32] [12]

2.1.4. Failures of TANF and the Case for Less Restrictive Welfare

The figure above illustrates a sharp and immediate drop in the number of families receiving federal welfare following the 1996 reform. In this respect, Clinton’s promise that it marked “the end of welfare as we know it” was true. However, TANF did not produce the long-term poverty reduction he had hoped for. Despite an initial decrease in poverty, poverty rates climbed for the next decade [32]. (Note that the 2019 estimates for poverty were biased downward due to

telephone-only polling during the pandemic, resulting in lower poverty reports – data between 2013 and 2017 were also estimates, which is why the graph above portrays them in gray[32].)

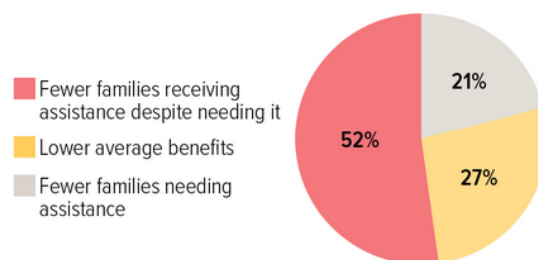
As TANF continues to exist, the number of families receiving assistance continues to drop, even as poverty rates remain stable [19]. Only about 23 of every 100 families whose incomes fall below the federal poverty line actually receive TANF assistance, and this assistance, in its severely limited and restrictive state, tends to be extremely unhelpful. States have tended to spend more of their TANF block grants on TANF work requirements at

the expense of direct payments to the recipients. TANF benefits for a family of three fall below 60% of the poverty line in *every* state and below 20% in 18 states. Furthermore, instead of reducing poverty, as Mead had argued, these work requirements actually caused a rise in deep poverty, as they did not help recipients find *quality* jobs – only unstable jobs characterized by below-poverty incomes. TANF time limits also spelled disaster for struggling families. Forty percent of families who were kicked off of TANF due to time limits had *no earnings*, and another 40% were making less than 50% of the poverty line[19]. It is worth mentioning that many experts feel that the US poverty line definition already dramatically undercounts poverty – many people above the poverty line still struggle to feed their children and remain housed [44].

What could a better system look like? Investing less money into means-testing – that is, the administrative and often humiliating and privacy-violating process of determining whether an applicant *deserves* welfare [24] – could leave more money to give to recipients. A 2011 study from the Center of Economic and Policy research found that implementing means-testing caused administrative costs to increase to a point where any money saved by having fewer families receive

Bulk of Cash Assistance's Decline Due to Worsened Access, Benefits — Not to Less Need

Decline in real value of total TANF/AFDC cash assistance (1993-2016) due to:



Note: TANF = Temporary Assistance for Needy Families; AFDC = Aid to Families with Dependent Children. TANF replaced AFDC in 1996. Families needing assistance were those whose incomes met their state's eligibility criteria for AFDC/TANF.

Source: Parolin, "Decomposing the Decline of Cash Assistance in the United States, 1993 to 2016"

CENTER ON BUDGET AND POLICY PRIORITIES | CBPP.ORG

Figure 3: Failures of TANF compared to AFDC [19]

benefits were negligible[28].

Furthermore, loosening the conditions of continuing to receive welfare could better help families get out of poverty. Cutting off TANF benefits after a time-limit only ensures that struggling families are left without any help, and requiring all TANF recipients to fulfill harsh work requirements in industries with no promise of advancement or transitioning into financial stability is exploitative and feeds a cycle of poverty. However, making welfare less conditional and less restrictive goes directly against much of the dominant conservative rhetoric of the past decades – it may be difficult to convince the public of such radical measures to help struggling people. If only we had an American case study involving direct welfare payments with no means-testing and no conditions that was received relatively well by conservative Americans that we could study the rhetoric around...

2.2. The Anomaly: COVID Stimulus Checks

From March of 2020 to the end of 2021, the federal US government distributed three rounds of Economic Impact Payments (EIP), more colloquially known as COVID-19 stimulus checks. These **direct cash payments** were available for every adult with an annual income lower than \$75,000, plus an extra \$500 for each child in the household, functioning similarly to AFDC. For Americans making more than \$75,000, the check value decreased gradually before finally becoming \$0 at any income above \$99,000. The accessibility and freedom of these funds far outclassed any welfare that TANF has been offering for the past few decades[38].

Ideologically, conservative Americans should have been as vehemently opposed to COVID stimu-

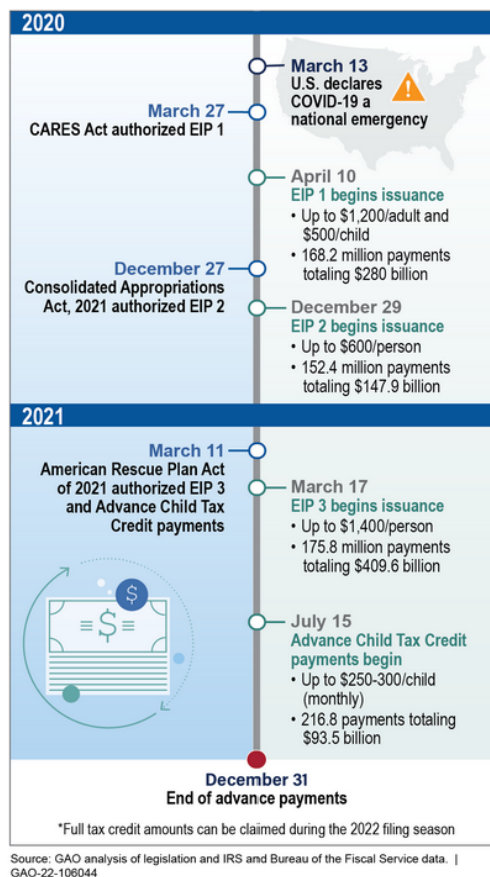


Figure 4: COVID Stimulus Check Timeline [38]

lus checks as they were to the late AFDC program, as both offered accessible direct-cash payments to struggling Americans. Yet, unexpectedly, such opposition did not come to fruition. Instead, what arose was a conservative sentiment that these stimulus checks were a necessity in a trying time. There was empathy given to those who were couldn't pay for rent or food because of the pandemic. Much of it could have been an effect of the extenuating circumstances, but there's potential in channeling this empathy for struggling folks to conditions outside the pandemic to advocate for similar mechanisms in our deeply flawed (and deteriorating) welfare state.

djohnmaga
28 December, 2020

I would normally consider myself as **fiscally conservative**. I would normally never agree with just doling out tax payer dollars in the form of stimulus checks, but **these are not normal times and the American people are in crisis and need the help of our government to feed their families and keep a roof over their heads** during this pandemic as it is unprecedented in history. Burgess is correct \$600 is a insult to the people and will do nothing to help. I see many people commenting that the things in the bill were in the Trump administrations budget and they are correct. but what they seem to fail to understand is that this was supposed to be a relief bill ! not a normal budget bill. Trump is fighting against packing this bill with things it was not meant for. Maybe the way bills are written and voted upon needs to be changed, one item bills one page bills one vote per bill. that way some things may actually get accomplished because when you pack a bill with many different items you will always get disagreement on some parts of it. if bills were one item each the good parts that everyone agrees to could be easily passed.

Reply · 90 · 4 · Share

Figure 5: Fox News Commenter 'djohnmaga' expressing their approval of the COVID stimulus checks

These stimulus checks were wholly unprecedented – the direct payments made by the government to Americans during the pandemic *far* surpassed the federal stimulus payments of the past.

Such a dramatic federal action using so many tax-payer dollars should have warranted a conservative outcry of government overreach, but the immediate pushback from conservative news outlets was not as strong as one may expect – after all, they too needed help and were receiving checks under the EIP payments. In fact, both progressive, liberal,

Stimulus payments in 2001, 2008-2009, and 2020-2021

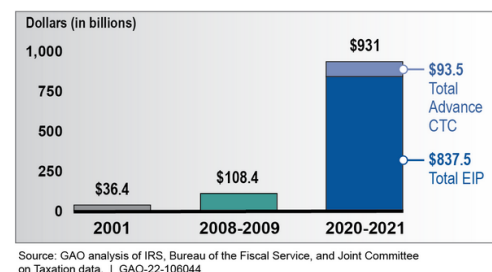


Figure 6: COVID Stimulus Check VS Past Stimulus Payments[38]

and conservative politicians alike were pushing the COVID stimulus check legislation through Congress in a rare show of bipartisanship[29]. This mass distribution of direct checks to both poor and middle class Americans proved the conservative fears of wastefulness and hedonism in poor populations to be unfounded – estimates on how Americans spent these checks found that poorer families tended to spend most of their check on necessities, such as rent, food, and other bills, while saving as much as they could. Further, these checks achieved many of the purported goals of welfare programs – they led to a *sharp* decrease in poverty, material hardship, food insufficiency, and even reported mental health symptoms, with more dramatic decreases from lower-income households[30]. A 2021 analysis of the COVID stimulus checks' effects writes:

"We believe the success of the federal government's relief measures may be due to the speed, breadth, and flexibility of its broad-based approach, primarily relying on cash transfers."

- Cooney and Shaefer, 2021 [30]

How can we recreate these conditions (minus the pandemic, of course) such that bipartisan support for poverty-reducing policies increases again? How do we sway conservative narrative around welfare towards the conservative narratives around these stimulus checks, which were significantly more positive, in order to build a broader base of support? In this paper, I take a computational approach to investigating and answering this question.

2.3. Computational Approaches to News Analysis: Past Work

Especially with the rise of the use of ML models in research, many past computational social science studies have employed sentiment analysis models to analyze political texts. Many papers use sentiment analysis and other computational methods to attempt to comprehend the vast amount of political information online, especially from social media platforms like Twitter [39]. Others, as Abercrombie and Batista-Navarro's 2019 literature review covered, focused on debate transcripts from Parliament, tracking position-taking using computational methods [25].

In regards to my topic, that is, framings of poverty and welfare programs in news media, a

wonderful paper by Rose and Baumgartner was published in 2013, investigating framings of the poor in US media coverage and resulting poverty policies from 1960 to 2008 [41]. This paper analyzed articles from 5 different news sources: the New York Times, Washington Post, Chicago Tribune, Baltimore Sun, and LA Times, and painted a nuanced picture of how poverty narratives shaped US poverty policy[41]. In contrast, my experiments aim to find effective rhetoric recommendations for policymakers in order to further my goal of reforming welfare programs such that help becomes more accessible for people in need. Because of this, I focus my analysis on an outwardly conservative news media like Fox News, and I look at COVID stimulus checks as a recent phenomenon where conservative narratives around help suddenly changed – using it as a case study for places of improvement in current welfare narratives. I also hope to patch a gap in current academic literature regarding the perception of COVID stimulus checks, as it was an unprecedented moment in US history, and the ways that political views shifted during this time were fascinating. However, I was unable to find substantial existing research on this topic.

3. Methodology

3.1. Scraping Fox News

Fox News was chosen as the site at which the analysis would be performed, as it is, by far, the most watched cable TV news platform. Furthermore, it is the most mainstream *conservative* news platform, home to many of the most popular conservative framings of recent news. As I was mostly interested in studying conservative narratives of welfare, especially since liberal political narratives have tended to follow their lead in recent years, I believed that focusing my dataset on Fox News articles and clip transcripts would yield more useful and applicable patterns for my analysis.

Using SerpAPI, I queried Google to find FoxNews links that mentioned welfare and related words, and repeated for COVID stimulus checks.

Welfare Query	site:foxnews.com inurl:news ("Social Security" OR "disability benefits" OR "unemployment benefits" OR "welfare") - inurl:video
COVID Stimulus Query	COVID stimulus checks site:foxnews.com

This yielded 2 .txt files with lists of Fox News links. The COVID stimulus check file contained 305 links, while the welfare program file contained 310 links.

Next, I scraped the content in each of these links by accessing each website using Python package 'requests', and using 'BeautifulSoup' to scrape the raw HTML from the website. I stripped this html to only include text between <p> and <h#> tags, which yielded all the text on the page. (I also scraped the date of each article and stored it in a separate category from the text.) I exported all of this information into a .json file. After manually combing through it, I wrote another program (clean.py) to clean the raw text to only include one copy of the actual article's contents (the raw text included other Fox News headlines on the same page, promotions for Fox News subscriptions, and other irrelevant legal disclaimers). This program also converted the scraped dates into machine-readable date strings.

This resulted in .json files containing all the article texts and dates that were now ready for analysis.

3.2. Go Sentiment Analysis

The model chosen for sentiment analysis was user joeddav's 'distilbert-base-uncased-go-emotions-student' [10], which is a classifier model trained on the dataset GoEmotions, a huge corpus owned and labeled by Google. GoEmotions contains over 58,000 Reddit comments that were labeled with 27 emotion categories, making it "the largest fully annotated English language fine-grained emotion dataset to date" [6]. The 27 emotions include:

- caring
- desire
- disapproval
- realization
- approval
- remorse
- curiosity
- excitement
- annoyance
- pride
- embarrassment
- grief
- optimism
- love
- surprise
- relief
- confusion
- anger
- fear
- disgust
- disappointment
- neutral
- nervousness
- admiration
- amusement
- joy
- sadness
- gratitude

The distilbert model was downloaded from huggingface onto Princeton University's adroit-vis computer cluster. A slurm script was submitted to run sentiment analysis using this model on both of the .json datasets from the COVID-19 stimulus check scraping and the welfare program scraping. The .csv files that it produced were downloaded locally for further analysis and investigation.

3.3. Frequency Analysis

In order to address the shortcomings of the sentiment analysis model results – that is, that the results could only identify the emotions in the texts, not the subject that they were directed at – I decided to perform a simple frequency analysis to find the words and subjects that were most commonly invoked alongside discussions. After finding the 200 words that were most often used across each corpus (I analyzed welfare and stimulus checks separately), I manually cleaned out any common stop words (e.g. 'and', 'but', 'it', 'the', etc.), and separated each remaining word into one of 6 categories:

- **Republican Administration**; words in this category explicitly referenced names and businesses affiliated with the Republican Administration – examples include Trump, Musk, Tesla, etc.
- **Democratic Administration**; words in this category explicitly referenced names and businesses affiliated with the Democratic Administration – examples include Pelosi, Biden, Harris, etc.
- **Security**; words in this category referenced law enforcement in some way – examples include ICE, police, security, etc.
- **Undeserving**; words in this category referenced some population that Fox News tended to characterize as contemptuous and undeserving of help – examples include immigrants, criminals, security, unemployed, fraud, etc.
- **Government**; words in this category referenced some system or institution of government – examples include court, house, senate, etc.
- **Economics**; words in this category referenced the economy in some way – examples include money, economy, business, stimulus, market, etc. These categories were color coded and put into word clouds for visualization.

3.4. Frame Dictionaries

However, these words and categories from the frequency analysis step were most helpful in helping me construct frame dictionaries. That is, lists of relevant and commonly used terms related to each of the 6 categories I created in the frequency analysis step. Then, in order to find which subjects were invoked most often in each article, I performed yet another frequency analysis, this time searching for the number of times each frame showed up in the article. Results were normalized so that the framings of longer articles weren't overrepresented, and both the welfare results and the COVID stimulus check results were graphed over time.

4. Results/Analysis

4.1. Go Sentiment Analysis

The original graphs for the results of the Go Sentiment Analysis plotted all 27 emotions on the same plane. As I found many of these emotions to be irrelevant, I focused my analysis on approval and disapproval metrics for both welfare and COVID stimulus checks.

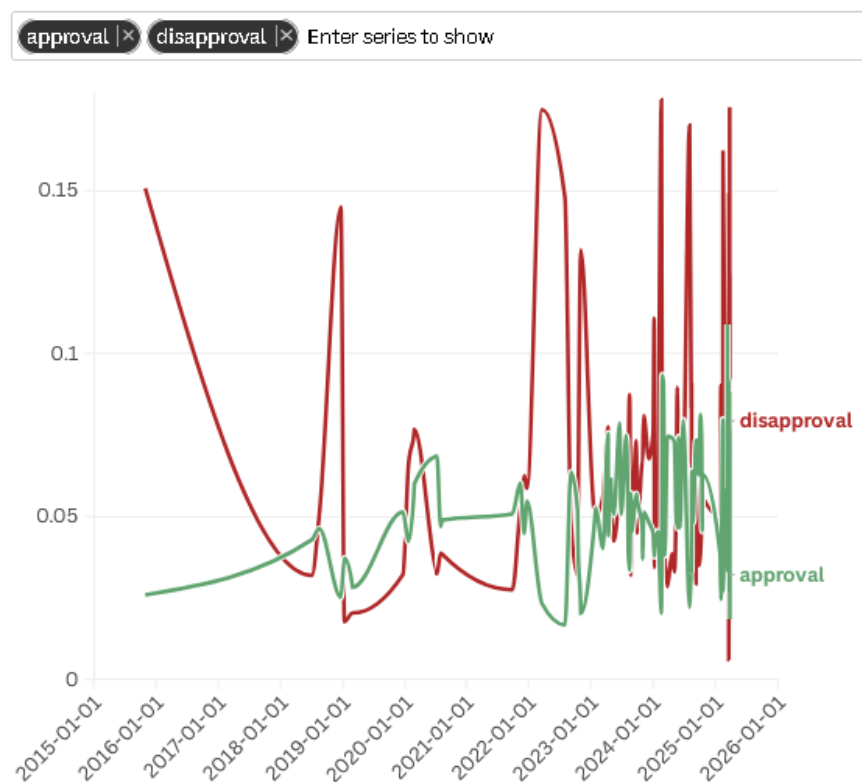


Figure 7: Disapproval/Approval Metrics for Fox News Coverage of Welfare

The figure above plots disapproval and approval metrics of the Fox News articles about welfare and social security programs over time. Disapproval overwhelmingly eclipses approval for almost every single data point, with a few exceptions from 2019 to 2022, coincidentally the same time period that COVID stimulus checks were distributed. Points are concentrated around more recent dates, likely due to search engine bias – as I used Google to query Fox News articles, Google likely favored returning more recent articles instead of older articles.

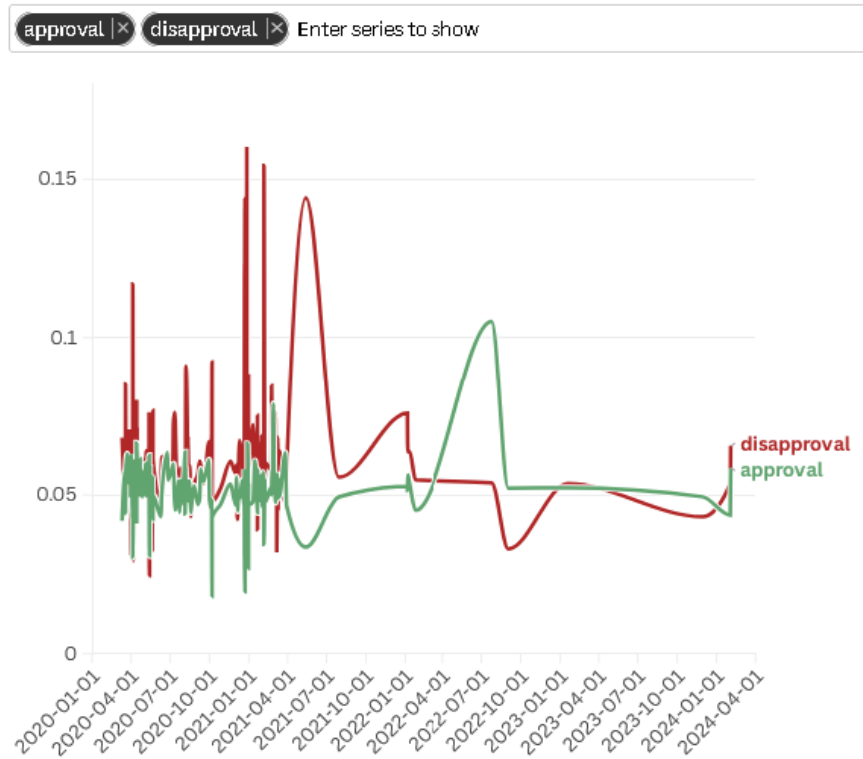


Figure 8: Disapproval/Approval Metrics for Fox News Coverage of COVID Stimulus Checks

This figure plots disapproval and approval metrics of the Fox News articles about COVID stimulus checks over time. Between 2020 and 2021, when COVID checks were first distributed, patterns in approval and disapproval mimic those of welfare programs – though there are points where approval and disapproval scores are close in value, disapproval still scores higher on almost every article. However, unlike the behavior of the welfare plot, COVID stimulus check dis/approval scores seem to flatten out over time, with approval scores tending to be higher than disapproval scores.

4.1.1. Case Study: Highest Disapproval Articles

In order to perform a more nuanced analysis of these plot behaviors, I'll be conducting a case study of the highest disapproval score articles from these results from both the welfare article list and the stimulus check article list. This is also a study into how the model identified disapproval and approval in articles. The following articles include the article from each list that had the highest positive difference in disapproval compared to approval.



Figure 9: Highest Disapproval Welfare Article[9]



Figure 10: Highest Disapproval COVID Stimulus Check Article [29]

Figure 11: Highest Disapproval Articles

The highest disapproval welfare article positions welfare in relation to a Des Moines suburban local government making a deal with Apple. The government gave Apple Inc. \$200 million in public subsidies in exchange for the company building a data center in Waukee that would give 50 new jobs to residents. The titular quote: "It's just welfare," came from a Des Moines citizen who testified that "Waukee probably would've been just fine without [the Apple data center]." In this way, the article reinforces existing narratives of welfare being both overly expensive and unnecessary. Other citizens in the article criticize that 50 jobs wasn't worth the price tag of \$200 million dollars in public subsidies, also linking this welfare-like deal with inefficiency and government corruption. Sen. Joe Bolcom even calls the financial benefits "giveaways to these super wealthy corporations," his criticism sounding remarkably similar to the "handouts" narratives of the Reagan-era. The high disapproval rating of this article can likely be attributed to the amount of times the text uses the word "critics," "unfair," and the critical quotes from the interviewees from Waukee [9].

The highest disapproval COVID stimulus check article was a Bernie Sanders category report that quoted Sanders' opposition to Trump's delay of the relief bill in December of 2020. Ultimately (and surprisingly), Trump and Sanders agree in their opinion that Americans deserve more money in the stimulus checks, but Sanders believed that funds (even if they were low) should get to struggling Americans as quickly as possible, while Trump refused to sign the legislation until payments were higher. Sanders called this delay "cruel," which likely contributed greatly to the high disapproval score of this article. However, this disapproval rating referred to Sanders' disapproval of Trump's decision, not the conservative public or the reporter's disapproval of COVID stimulus checks. In fact, most of the comments under this article chime in support of increasing COVID stimulus check payments, but blame Democrats and progressives for their delay[29].

4.1.2. Case Study: Highest Approval Articles

Similarly, I'll also study the articles that had the highest positive approval scores in relation to disapproval.



Figure 12: Highest Approval Welfare Article [43]

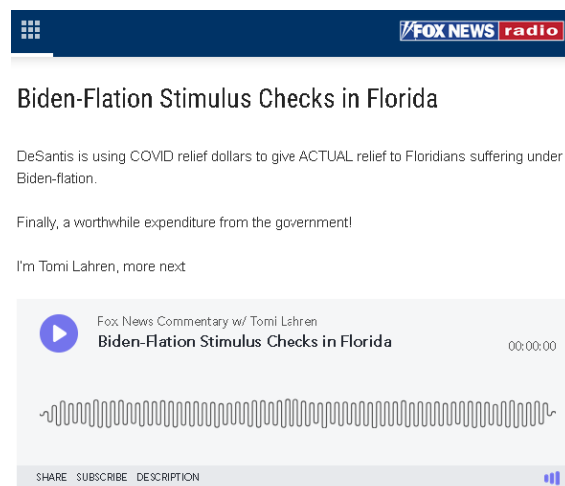


Figure 13: Highest Approval COVID Stimulus Check Article [36]

Figure 14: Highest Approval Articles

The highest approval welfare article reported that the additional \$600 that was added to unem-

ployment welfare benefits during the COVID pandemic was being removed, as economists claimed that the additional money was no longer needed. In the article, familiar Republican arguments against welfare were also raised:

Republican lawmakers, though, had ramped up warnings that the boost in jobless benefits amid the pandemic would “push unemployment higher,” as many individuals are able to collect more money through the unemployment programs than they made while on the job.

It's not entirely clear where the higher approval rating for this article came from, as the rhetoric invoked in it is remarkably similar to many of the other Fox News welfare articles, though there is a smaller focus on "illegals," "migrants," "criminals," and other "undeserving" populations compared to other articles. Comments below this welfare article are mixed; some commenters support sustaining the additional payments as their economics have been hit hard by COVID, and others say that "\$600 a week was ridiculous from the start. Many of the unemployed will never want to work again receiving that kind of money for simply not working" [43].

The highest approval COVID stimulus check article was a short transcript of a clip from a Fox-affiliated podcast: "Fox News Commentary w/ Tomi Lahren." In this transcript, Lahren praises Florida Republican governor Ron DeSantis for "using COVID relief dollars to give ACTUAL relief to Floridians suffering under Biden-flation." Her commentary connects stimulus checks very closely to Governor DeSantis, and focuses largely on how these stimulus payments will benefit *real, American families*, an unmistakable divergence from Fox News' coverage of welfare, which in contrast benefit illegal, un-American criminals[36].

4.2. Frequency Analysis Word Clouds

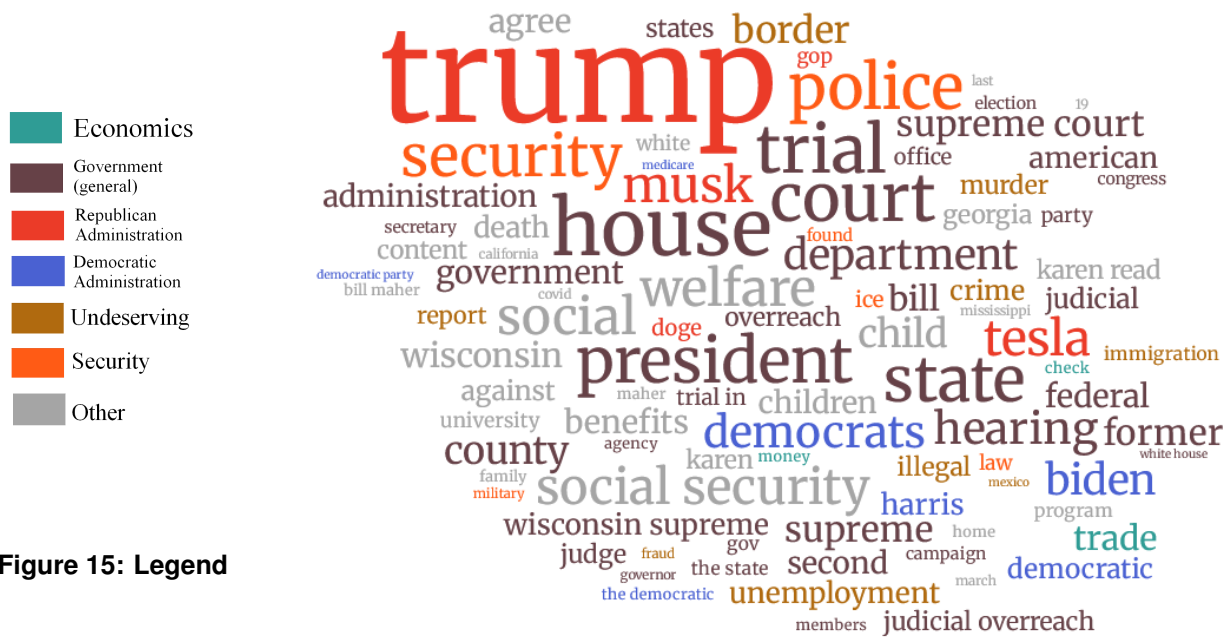


Figure 15: Legend

Figure 16: Welfare Article Word Cloud

Figure 17

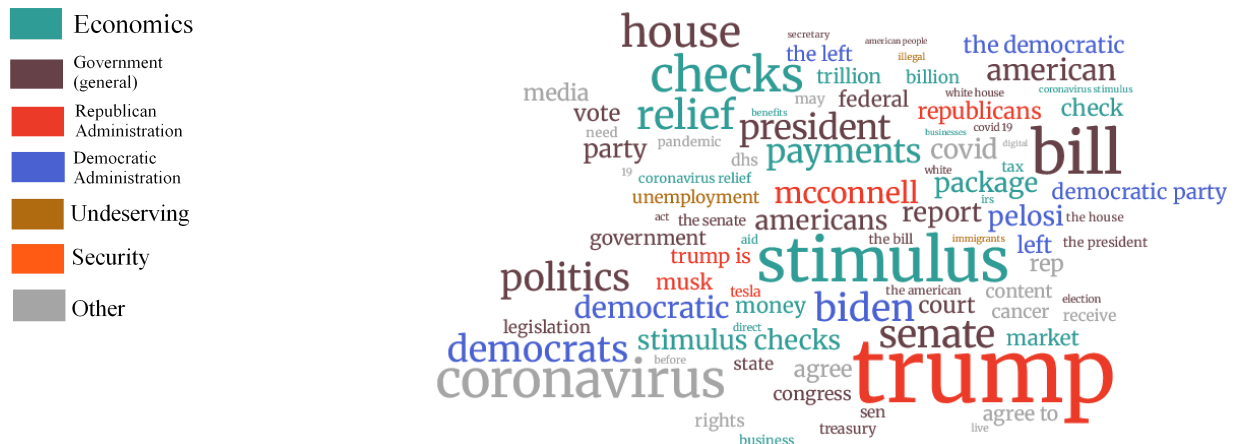


Figure 18: Legend (again)

Figure 19: COVID Stim Check Article Word Cloud

Figure 20

These word clouds were largely intermediate results for the building of the frame dictionaries in the next step, but they still contain some valuable insights. Visually, we can observe that welfare articles tend to tie much more closely towards the subjects of Government (general), Undeserving, and Security (the terms "Police" and "Security" are large). Meanwhile, COVID Stimulus check articles tend to tie closer to economics and the Democratic party.

4.3. Framing Analysis

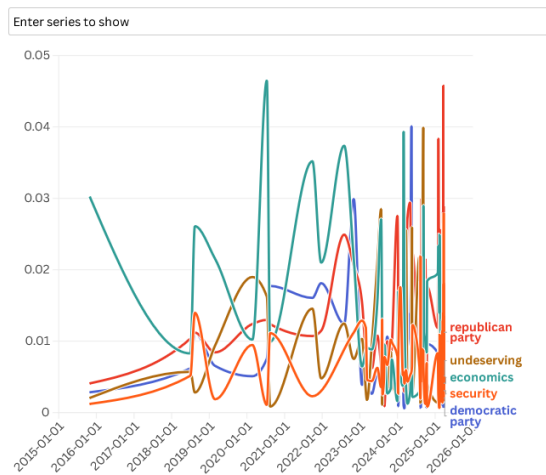


Figure 21: Welfare Frame Analysis

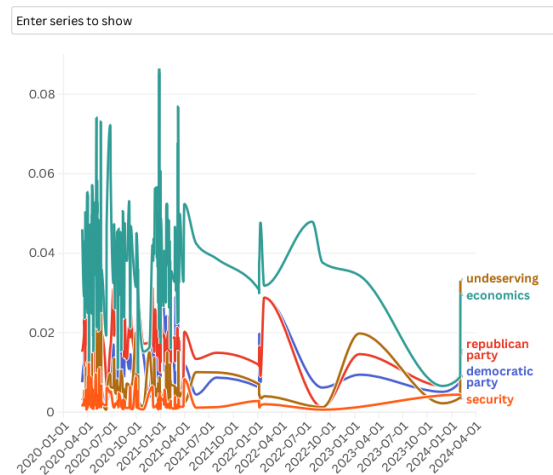


Figure 22: COVID Stim Check Frame Analysis

Figure 23

As observed in figures 20 and 21, the frames are generally evenly applied throughout the welfare articles, though the economics frame is mentioned more than other frames by a small margin that becomes moot after 2023. On the other hand, the economics frame consistently *dominates* the articles on COVID stimulus checks by a wide margin. This may be in part due to their name including both the words "stimulus" and "checks," which were part of the frame dictionary for economics, but they were also often called by other names in the articles.

Discussions of undeserving populations were also markedly lower in the COVID stimulus Check frame analysis compared to the Welfare analysis.

5. Discussion

There is a marked difference in the way that Fox News speaks about government welfare programs versus the way that they speak about COVID stimulus checks, though the two do overlap. During the time when COVID stimulus checks were first being distributed, conservative welfare narratives seem to have been affected by the COVID stimulus check narratives that were being published concurrently, which made approval ratings for federal welfare programs become significantly higher in 2020-2022. Perhaps the conservative public, experiencing conditions like unemployment and financial troubles due to factors out of their control (such as the COVID pandemic), became more amenable to federal welfare programs because they finally saw the worth of a social safety net when they were falling themselves. After all, unemployment shot up during COVID after mass layoffs and shutdowns – it seems to be in many Fox News commenters' stories that they, who had worked hard for their country at their jobs for decades, found applying to welfare programs such as unemployment benefits or TANF deeply lacking.

However, this quickly shifted back into Reagan-like rhetoric that painted welfare recipients as unworthy and the welfare programs that served them as excessively generous and foolish. This rhetoric was also accompanied by a new focus on "illegals" – that is, illegal immigrants, usually from Mexico, usually characterized as drug-dealing criminals associated with the Mexican cartel. Many Fox News articles focused on government benefits going to this (imagined) population in a remarkably similar pattern to Reagan's use of Black single mothers as welfare queens.

The primary difference between Fox News' coverage of welfare and COVID stimulus checks, found both in the frequency analysis and the framing analysis, was the focus on the economy when talking about COVID stimulus checks. These checks were framed as a saving grace, not just to the American public, but to the American *economy*, as giving people spendable money allowed for more capital flow. As eyes turned to the economy as a metric of success of the COVID stimulus checks, people across the isle saw both poverty and material hardship fall sharply. This same privilege is not granted to Fox News narratives on federal welfare programs however – these narratives tend to

center the morality of welfare recipients as the priority, instead of the effects of the welfare programs on the forces they are trying to combat – that is, unemployment, poverty, and material hardship. One can observe this in Figure 20 and 21, where the undeserving frame for welfare articles is, on average, higher than the same frame for COVID stimulus check articles.

5.1. Conclusions and Recommendations

Based on the results of this research, it seems that conservatives tend to react better to federal help programs when they are framed in terms of their benefits to the economy, with less emphasis on characterizations of the recipients of the help. Furthermore, I suspect that insight into the experience of a unemployment/TANF/disability benefits recipient – that is, exposure to the humiliation of means-testing for a (sometimes, below) bare minimum amount of financial help during the COVID pandemic – boosted the general American public’s empathy for welfare recipients, which resulted in significantly more positive perceptions of welfare programs during the COVID pandemic.

Thus, in order to boost public perceptions of welfare programs for the ultimate goal of reducing means-testing and increasing direct-cash payments for welfare programs, I tentatively propose two changes in rhetoric:

1. Focusing on welfare programs’ resulting boosts to the economy, rather than making the programs a moral issue.
2. Capitalize on newfound empathy from conservatives from the COVID-19 pandemic towards welfare recipients – show the humiliation and privacy violation of means-testing, as well as how a lifestyle funded by welfare payments actually looks (below the poverty line)

5.2. Limitations

As briefly mentioned in the Results and Analysis section, search engine bias (most evidently seen in the welfare article dataset’s bias favoring recent articles over older ones) was present in both the welfare article dataset and the COVID stimulus check dataset. This resulted in datasets that favored articles that were more recent and had more views, possibly skewing the dataset towards the more dramatic and controversial articles.

Partly because of search engine bias, there was an uneven number of articles across time, which made the graphs of COVID stimulus check scores and welfare program scores hard to compare, as they had concentrated data points in different places across time. Furthermore, there exists limitations in my use of Fox News as a representative for conservative news media – Fox News may not capture all the nuances of conservative beliefs. Also, Fox News is primarily a cable news channel rather than an article writing body, and there were not as many transcripts of broadcasts as I expected in my scraping, which may also contribute to inaccuracies in my datasets in capturing general conservative rhetoric.

Additionally, the Go Sentiment analysis model I used was trained primarily on Reddit comments, which tend to be more casual in tone than news articles, which may have resulted in inaccuracies in the sentiment analysis results. I was also unable to confirm if every single scraped link in each dataset was relevant – there is a chance that some of the scraped links in each dataset were not actually related to welfare programs or COVID stimulus checks, resulting in some irrelevant data in my plots.

5.3. Future Work

I found, in my case studies, that the Fox News comments were an invaluable data source for analyzing conservative reactions to COVID stimulus check or welfare program updates. In future work, I would highly recommend analyzing these comments alongside article text (I had made an attempt to scrape comments near the beginning of my experiments, but I was unable to find an effective way to scrape them). In order to supplement the datasets' difference in number of articles over time as well as draw more focus to how narratives of each affected the other, I'd also highly recommend analyzing this data through 2 or 3 year timespans. In order to make a more sound quantitative analysis, I think that comparing randomly chosen 2-year increments from both these datasets would also be worthwhile.

To get a more rounded look at conservative narratives around government help for those in poverty, it would be worthwhile to analyze other online conservative spaces, like Truth Social or

QAnon. Additionally, future work could use an LLM to find less concrete types of frames, such as George Lakoff's proposed framework, arguing that progressives tend to use a "nurturant parent" frame, while conservatives opt for a "strict father" frame [26].

References

- [1] Beautiful Soup Documentation — Beautiful Soup 4.4.0 documentation. [Online]. Available: <https://beautiful-soup-4.readthedocs.io/en/latest/>
- [2] Clinton Says He Will Sign Welfare Overhaul; House Passes It. [Online]. Available: <https://web.archive.org/web/20200807202843/https://www.apnews.com/f11a3d867b896908c6c598e31fb94ff8>
- [3] The Fed - Acts of Congress and COVID-19: A Literature Review on the Impact of Increased Unemployment Insurance Benefits and Stimulus Checks. [Online]. Available: <https://www.federalreserve.gov/econres/notes/feds-notes/acts-of-congress-and-covid-19-unemployment-insurance-benefits-and-stimulus-checks-20210224.html>
- [4] Frame the Debate: Insights from Don't Think of an Elephant! - The Commons. [Online]. Available: <https://commonslibrary.org/frame-the-debate-insights-from-dont-think-of-an-elephant/>
- [5] “Frances Fox Piven.” [Online]. Available: https://en.wikipedia.org/w/index.php?title=Frances_Fox_Piven&oldid=1284109833
- [6] GoEmotions: A Dataset for Fine-Grained Emotion Classification. [Online]. Available: <https://research.google/blog/goemotions-a-dataset-for-fine-grained-emotion-classification/>
- [7] Impact of the Coronavirus Stimulus Checks on the Economy - Julian Samora Research Institute - Michigan State University. [Online]. Available: <https://jsri.msu.edu/publications/nexo/vol-xxv/no-1-fall-2021/impact-of-the-coronavirus-stimulus-checks-on-the-economy>
- [8] Implementing Web Scraping in Python with BeautifulSoup | GeeksforGeeks. [Online]. Available: <https://www.geeksforgeeks.org/implementing-web-scraping-python-beautiful-soup/>
- [9] Iowans respond to delayed project giving Apple \$200 million in public incentives: ‘It’s just welfare’ | Fox News. [Online]. Available: <https://www.foxnews.com/us/iowans-respond-to-delayed-project-giving-apple-200-million-in-public-subsidies-its-just-welfare>
- [10] Joeddav/distilbert-base-uncased-go-emotions-student · Hugging Face. [Online]. Available: <https://huggingface.co/joeddav/distilbert-base-uncased-go-emotions-student>
- [11] “Lawrence Mead.” [Online]. Available: https://en.wikipedia.org/w/index.php?title=Lawrence_Mead&oldid=1273426965
- [12] LIST OF TABLES - Administration for Children and Families. [Online]. Available: https://www.congress.gov/crs_external_products/R/HTML/R48413.web.html
- [13] The New Politics of Poverty: The Non-Working Poor in America. City Journal. [Online]. Available: <https://www.city-journal.org/article/the-new-politics-of-poverty-the-non-working-poor-in-america>
- [14] Poverty is the 4th greatest cause of U.S. deaths | UCR News | UC Riverside. [Online]. Available: <https://news.ucr.edu/articles/2023/04/17/poverty-4th-greatest-cause-us-deaths>
- [15] Project 100 research | GiveDirectly. [Online]. Available: <https://www.givedirectly.org/project-100-research/>
- [16] Q&A: Charles Murray; Of Babies And Stick - The New York Times. [Online]. Available: <https://www.nytimes.com/1986/04/11/us/q-a-charles-murray-of-babies-and-stick.html>
- [17] Ronald Reagan on Welfare & Poverty. [Online]. Available: https://www.ontheissues.org/celeb/Ronald_Reagan_Welfare+_Poverty.htm
- [18] Stimulus Checks: Direct Payments to Individuals during the COVID-19 Pandemic | U.S. GAO. [Online]. Available: <https://www.gao.gov/products/gao-22-106044>
- [19] TANF Cash Assistance Helps Families, But Program Is Not the Success Some Claim | Center on Budget and Policy Priorities. [Online]. Available: <https://www.cbpp.org/research/income-security/tanf-cash-assistance-helps-families-but-program-is-not-the-success-some>
- [20] Trends and Facts on Cable News | State of the News Media. [Online]. Available: <https://www.pewresearch.org/journalism/fact-sheet/cable-news/>
- [21] U.S. Poverty Statistics and Facts (Updated for 2025). Debt.org. [Online]. Available: <https://www.debt.org/faqs/americans-in-debt/poverty-united-states/>
- [22] War on Poverty | History, Speech, Significance, & Facts | Britannica. [Online]. Available: <https://www.britannica.com/topic/War-on-Poverty>
- [23] What Americans Think about Poverty, Wealth, and Work. Cato Institute. [Online]. Available: <https://www.cato.org/publications/survey-reports/what-americans-think-about-poverty-wealth-work>
- [24] What is means testing and the case against using it | Vox. [Online]. Available: <https://www.vox.com/2021/10/15/22722418/means-testing-social-spending-reconciliation-bill>
- [25] G. Abercrombie and R. Batista-Navarro, “Sentiment and position-taking analysis of parliamentary debates: A systematic literature review,” vol. 3, no. 1, pp. 245–270. [Online]. Available: <http://arxiv.org/abs/1907.04126>
- [26] Antje. Frame the Debate: Insights from Don't Think of an Elephant! The Commons. [Online]. Available: <https://commonslibrary.org/frame-the-debate-insights-from-dont-think-of-an-elephant/>

- [27] B. c. A. U. April 5 and 2024. Aid to Families with Dependent Children. Benefits.com - We Make Government Benefit Program Information Easier to Understand. [Online]. Available: <https://benefits.com/glossary/aid-to-families-with-dependent-children/>
- [28] D. Baker and H. J. Rho, "The Potential Savings to Social Security from Means Testing."
- [29] R. Blitzer. Sanders calls Trump delay on relief bill 'cruel,' while agreeing with push for \$2,000 payments. Fox News. [Online]. Available: <https://www.foxnews.com/politics/sanders-trump-delay-coronavirus-relief-bill-cruel-stimulus-checks>
- [30] P. Cooney and H. L. Shaefer, "MATERIAL HARDSHIP AND MENTAL HEALTH FOLLOWING THE COVID-19 RELIEF BILL AND AMERICAN RESCUE PLAN ACT."
- [31] W. Copeland, *And The Poor Get Welfare: The Ethics of Poverty in the United States* (Churches Center for Theology and Public Policy).
- [32] J. Dalaker, "Poverty in the United States in 2019."
- [33] S. Danziger and R. Haveman, "The Reagan administration's the poor."
- [34] E. Falcettoni and V. Nygaard, "Acts of Congress and COVID-19: A Literature Review on the Impact of Increased Unemployment Insurance Benefits and Stimulus Checks," vol. 2021.0, no. 2839. [Online]. Available: <https://www.federalreserve.gov/econres/notes/feds-notes/acts-of-congress-and-covid-19-unemployment-insurance-benefits-and-stimulus-checks-20210224.htm>
- [35] J. Madrick. Testimony: How We Measure Poverty Is Failing Americans. The Century Foundation. [Online]. Available: <https://tcf.org/content/commentary/testimony-measure-poverty-failing-americans/>
- [36] K. Michael. Biden-Flation Stimulus Checks in Florida. FOX News Radio. [Online]. Available: <https://radio.foxnews.com/2022/07/25/biden-flation-stimulus-checks-in-florida/>
- [37] migrate. Did Stimulus Checks Save the U.S. Economy? The National Interest. [Online]. Available: <https://nationalinterest.org/blog/reboot/did-stimulus-checks-save-us-economy-186800/>
- [38] U. S. G. A. Office. Stimulus Checks: Direct Payments to Individuals during the COVID-19 Pandemic | U.S. GAO. [Online]. Available: <https://www.gao.gov/products/gao-22-106044>
- [39] A. Ramesh, "Algorithmic Analysis of Political Tweets: A Systematic Literature Review," pp. 185–195. [Online]. Available: <https://srels.org/index.php/sjim/article/view/170345>
- [40] U. o. C. Riverside. Poverty is the fourth greatest cause of US deaths, analysis finds. [Online]. Available: <https://medicalxpress.com/news/2023-04-poverty-fourth-greatest-deaths-analysis.html>
- [41] M. Rose and F. R. Baumgartner, "Framing the Poor: Media Coverage and U.S. Poverty Policy, 1960–2008," vol. 41, no. 1, pp. 22–53. [Online]. Available: <https://onlinelibrary.wiley.com/doi/full/10.1111/psj.12001>
- [42] S. Silver. Why Did Trump Insist upon His Name Being on the First Stimulus Checks? - The National Interest. [Online]. Available: <https://nationalinterest.org/blog/reboot/why-did-trump-insist-upon-his-name-being-first-stimulus-checks-186539>
- [43] B. Singman. Additional \$600 a week in jobless benefits about to expire. Fox News. [Online]. Available: <https://www.foxnews.com/politics/additional-600-a-week-in-jobless-benefits-about-to-expire>
- [44] C. Thornton. Decades-old US poverty level formula 'makes no sense' in 2022, experts say. Here's why it's still used. USA TODAY. [Online]. Available: <https://www.usatoday.com/story/news/nation/2022/12/09/why-federal-poverty-line-not-effective/10827076002/>

6. Appendix

6.1. Programs

6.1.1. Scraping Program (example below is of welfare articles from Fox News)

```
from serpapi import GoogleSearch

import itertools

# queries serpAPI for fox news articles about 'welfare', created welfare_fox_links.txt
def google_search(query, max_results=500):
    links = []
    for start in range(0, max_results, 100):
        params = {
            "q": query,
            "num": 100,
            "start": start,
            "api_key": "4ceb3ea41d4b4aedc75aed5b6eabeeb7fed55bee9bec5630e287430b03d81b"
        }
        search = GoogleSearch(params)
        results = search.get_dict()
        page = [r["link"] for r in results.get("organic_results", [])] \#if "foxnews.c
        links.append(page)
    return links

# Run search

# edit search to find articles from foxnews
query = "site:foxnews.com inurl:news (\"Social Security\" OR \"disability benefits\" O
```

```
fox_news_links = google_search(query, max_results=500)

# Flatten the list of lists into a single list
fox_news_links = list(itertools.chain.from_iterable(fox_news_links))
print("\n".join(fox_news_links))
```

6.1.2. Cleaning Article Text: clean.py

```
'''
Cleans up unnecessary text from Fox News articles that were scraped by BeautifulSoup.
'''

import re
import sys
import json

junk_phrases = [
    "Legal Statement. Mutual Fund and ETF data provided by"
]

# Remove lines that contain junk phrases
def remove_junk_lines(text):
    lines = text.splitlines()
    clean_lines = [
        line for line in lines
        if not any(phrase.lower() in line.lower() for phrase in junk_phrases)
    ]
    return "\n".join(clean_lines).strip()

# Remove duplicate paragraphs
def remove_duplicate_paragraphs(text):
```

```

paragraphs = text.split("\n")

seen = set()

unique_paragraphs = []

for para in paragraphs:

    stripped = para.strip()

    if stripped and stripped not in seen:

        unique_paragraphs.append(stripped)

        seen.add(stripped)

return "\n".join(unique_paragraphs)

# clean function

def fully_clean(text):

    text = remove_junk_lines(text)

    text = remove_duplicate_paragraphs(text)

    return text

# open JSON

with open(sys.argv[1], 'r', encoding='utf-8', errors='ignore') as f:

    data = json.load(f)

# clean each text field

for article in data:

    if "text" in article:

        article["text"] = fully_clean(article["text"])

# save as json

with open(sys.argv[1][: -5] + "_cleaned.json", 'w') as f:

    json.dump(data, f, indent=4)

```

```
print(f"Cleaned {len(data)} articles and saved to {sys.argv[1][:5]}_cleaned.json")
```

6.1.3. Sentiment Analysis Code

```
import pandas as pd

import nltk

import torch

from bs4 import BeautifulSoup

from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer

import json

import sys

from transformers import pipeline, AutoModelForSequenceClassification, AutoTokenizer

from datasets import Dataset


# Initialize sentiment analyzer

analyzer = SentimentIntensityAnalyzer()


model_path = "/home/kr5379/IW-Programming-1/go_emotions_model"

# Load the model and tokenizer offline

device = "cuda" # if torch.cuda.is_available() else "cpu"

model = AutoModelForSequenceClassification.from_pretrained(model_path)

tokenizer = AutoTokenizer.from_pretrained(model_path)


# Initialize pipeline with local model

emotion_analyzer = pipeline("text-classification",

                             model=model,

                             tokenizer=tokenizer,

                             top_k=None,

                             truncation=True,
```

```

        batch_size=16,

        device=0)

# Read json file
with open(sys.argv[1], "r") as f:
    articles = json.load(f)

# Step 1: Prepare data for the Dataset
emotion_inputs = [{"text": article["text"][:512]} for article in articles]
emotion_dataset = Dataset.from_list(emotion_inputs)

# Convert the dataset to a list of strings (the "text" column only)
emotion_inputs = [item['text'] for item in emotion_dataset]

# Step 2: Run emotion analysis efficiently
print(emotion_dataset[0])
emotion_results = emotion_analyzer(emotion_inputs, batch_size=16)

# Step 3: Combine results
data = []

for article, emotions in zip(articles, emotion_results):
    text = article["text"]
    sentiment = analyzer.polarity_scores(text)
    emotion_dict = {emo['label']: emo['score'] for emo in emotions}

    data.append({
        "URL": article['url'],
        "Text": text[100:500], # Snippet
        "Sentiment Score": sentiment["compound"],

```

```

        "Date": article['date'],

        **emotion_dict

    })

# Save results

df = pd.DataFrame(data)

df.to_csv(sys.argv[1][:5] + ".csv", index=False)

```

6.1.4. Frequency Analysis

```

from sklearn.feature_extraction.text import TfidfVectorizer

import json

import numpy as np

import pandas as pd

# load the covid json file

# FOR COVID

# with open('covid_articles_cleaned.json', 'r') as f:

#     data = json.load(f)

# Extract the text fields

# texts = [article["text"] for article in data if "text" in article]

# FOR WELFARE

with open('welfare-1_cleaned.json', 'r') as f:

    data1 = json.load(f)

with open('welfare-2_cleaned.json', 'r') as f:

    data2 = json.load(f)

# Extract the text fields

texts1 = [article["text"] for article in data1 if "text" in article]

```

```

texts2 = [article["text"] for article in data2 if "text" in article]
texts = texts1 + texts2

vectorizer = TfidfVectorizer(max_df=0.9, min_df=5, ngram_range=(1, 2))
X = vectorizer.fit_transform(texts) # or stimulus_texts
feature_names = vectorizer.get_feature_names_out()

frequencies = X.toarray().sum(axis=0) # total frequency for each term
freq_dict = dict(zip(feature_names, frequencies))

# sort + get top N
top_n = 200
sorted_terms = sorted(freq_dict.items(), key=lambda x: x[1], reverse=True)[:top_n]
df = pd.DataFrame(sorted_terms, columns=["term", "frequency"])
df.to_csv("welfare_term_frequencies.csv", index=False)

```

6.1.5. Frame Analysis

```

import re

from collections import Counter

import json

import sys

import pandas as pd

frame_dict = {

    "democratic party":[

        "biden", "democrats", "democratic", "pelosi",

        "obama", "clinton", "hillary", "senate",

        "congress", "the left", "progressive", "medicare"

    ],

```

```

"republican party":[
  "trump", "republicans", "republican", "mcconnell",
  "conservative", "gop", "right-wing",
  "congress", "the right", "conservatives",
  "musk", "doge", "elon", "maga",
  "reagan", "libertarian", "libertarians",
  "liberty", "liberties", "libertarianism",
  "tesla"
],
"undeserving":[
  "unemployment", "welfare", "welfare queen",
  "welfare queens", "welfare state", "welfare states",
  "welfare dependency", "welfare dependents",
  "fraud", "illegal", "immigrants",
  "crime", "border", "mexico", "mexican",
  "report", "immigration", "illegals",
  "criminal", "criminals", "criminality",
  "murder", "lazy", "laziness", "lazy people",
  "lazy people", "entitlement", "entitlements",
  "entitled", "entitlement programs", "moral",
  "irresponsible"
],
"economics":[
  "economy", "economic", "inflation",
  "interest", "interest rates", "interest rate",
  "debt", "deficit", "taxes",
  "tax", "taxpayer", "taxpayers",

```

```

    "spending", "spend", "spending cuts",
    "spending cut", "budget", "budgets",
    "financial", "finance", "trade", "money",
    "market", "markets", "stock", "check", "checks",
    "stimulus", "relief", "payments", "package",
    "stimulus package", "stimulus payments",
    "stimulus check", "stimulus checks", "trillion",
    "billion", "business", "businesses",
    "corporation", "corporations", "corporate",
    "aid", "direct", "irs", "benefits"
],
"security":[
    "police", "security", "ice", "border patrol",
    "national guard", "military", "army",
    "navy", "air force", "armed forces",
    "law enforcement", "law enforcement agencies",
    "law", "laws", "legal", "military", "found"
],
}

```

```

# makes all text lowercase and removes punctuation

```

```

def preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text)
    return text

```

```

# counts the number of times each frame appears in text
def detect_frames(text, frame_dict):
    counts = Counter()
    total_words = len(text.split())
    for frame, keywords in frame_dict.items():
        for keyword in keywords:
            # You can match whole words only
            if re.search(rf'\b{re.escape(keyword)}\b', text):
                counts[frame] += len(re.findall(rf'\b{re.escape(keyword)}\b', text))/t
    return counts

results = []

with open(sys.argv[1], 'r', encoding='utf-8', errors='ignore') as f:
    articles = json.load(f)
for article in articles:
    text = preprocess(article['text'])
    frame_counts = detect_frames(text, frame_dict)
    results.append({
        "URL" : article['url'],
        "frame_counts": frame_counts
    })

# Convert the results to a flat structure suitable for CSV
df = pd.DataFrame([
    {
        "URL": r["URL"],
        **r["frame_counts"]
    }
])

```

```

    } for r in results

])

# save to csv

df.to_csv(sys.argv[1][:5] + "_frames.csv", index=False)

```

6.1.6. Github Repository

All of the other code that I wrote is in this public repository:

<https://github.com/Katypus/IW-Programming>

6.2. Interactive Flourish Graphs

Interact with my Flourish graphs + the data used to make them at these links!

1. [Welfare Sentiment Analysis Graph](#)
2. [COVID Stimulus Checks Sentiment Analysis Graph](#)
3. [Welfare Article Word Cloud](#)
4. [COVID Stimulus Check Article Word Cloud](#)
5. [Welfare Framing Analysis Graph](#)
6. [COVID Stimulus Check Framing Analysis Graph](#)

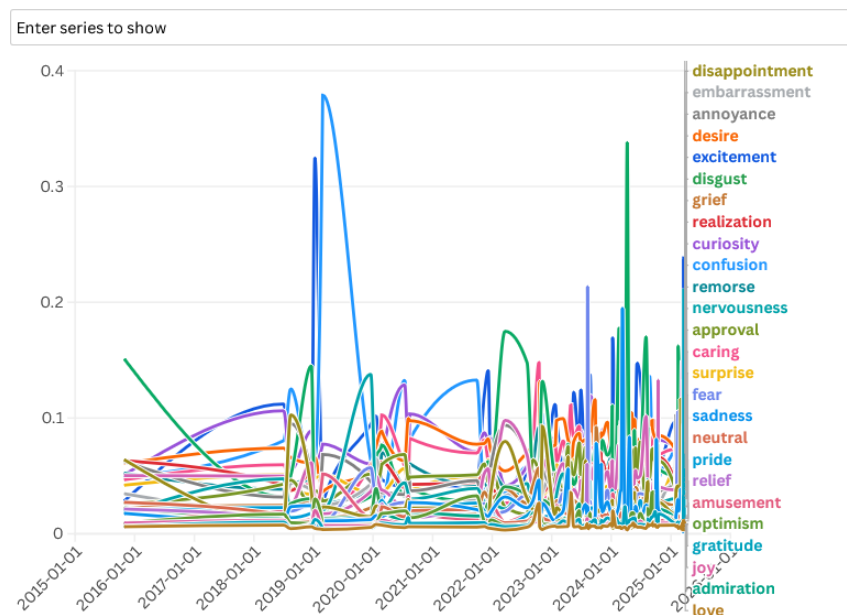


Figure 24: Full Sentiment Analysis Results for Fox News Coverage of Welfare

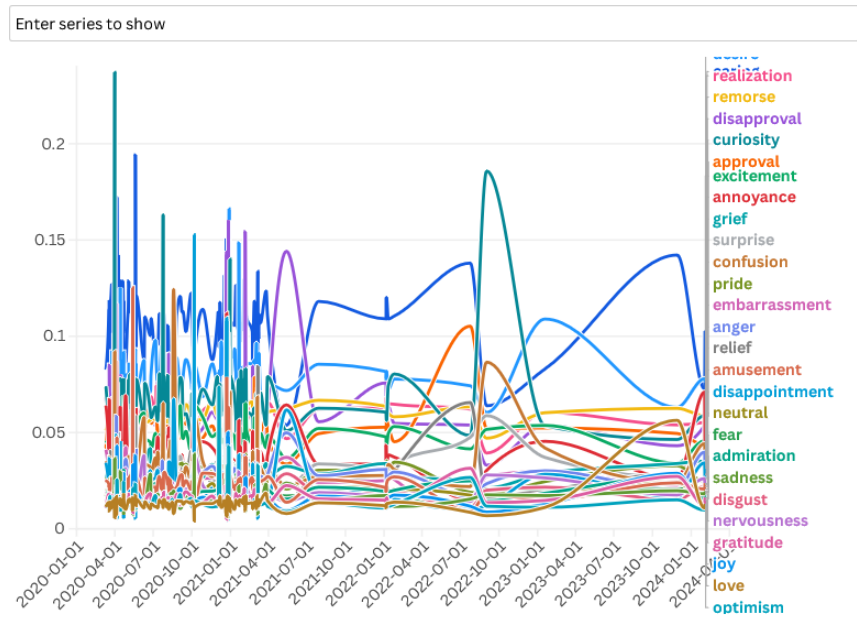


Figure 25: Full Sentiment Analysis Results for Fox News Coverage of COVID Stimulus Checks